

DFA/SFIG-EHOA: Regime-Aware Stability Feedback and Confidence-Gated Interaction Guidance for Feature Selection

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Abstract

Enhanced Hiking Optimization Algorithm (EHOA) is a recent wrapper feature selector that combines chaotic initialization, a dynamic sweep factor, and a binary transfer. Its search dynamics, however, do not react to the reproducibility of selected features under data perturbation. This paper introduces DFA/SFIG-EHOA, a modular extension with (i) a regime-aware controller that maps cross-resampling stability, population diversity, progress, and stagnation to bounded sweep and inertia adjustments, and (ii) a binary transition guided by entropy-gated feature reliability and a training-only relevance/redundancy signal. Instability alone is not allowed to trigger more exploration, preventing a self-reinforcing instability loop. The baseline fitness is unchanged. A paired 2-by-2 ablation was conducted on Breast Cancer Wisconsin and Wine with ten fixed seeds and repeated stratified hold-outs. The full method changed mean balanced accuracy by -0.0056 and 0.0000 relative to EHOA, while improving chance-corrected Nogueira stability by 0.0726 and 0.0365. On Breast Cancer it also selected 1.4 fewer features. No Holm-corrected predictive comparison was significant, and runtime was approximately three times the baseline. In the three-objective outcome space (maximize balanced accuracy and stability; minimize feature count), the full method is Pareto non-dominated on both datasets and dominates SF-EHOA on Wine. The evidence therefore supports a scoped stability-reduction trade-off, not universal predictive superiority. All 80 runs, traces, configurations, tests, figures, and statistical outputs are released with the implementation.

Keywords: feature selection; stability; metaheuristic; Hiking Optimization Algorithm; interaction guidance; reproducibility

1 Introduction

Feature selection (FS) seeks a compact subset that preserves predictive signal. Wrapper FS is attractive because it evaluates subsets using the downstream learner, but its combinatorial search is expensive and stochastic subsets may be unstable. In medical or scientific use, instability matters because small changes to the training sample can produce different explanatory features [6].

Hegazy et al. proposed EHOA for accurate and interpretable FS [4]. EHOA improves the Hiking Optimization Algorithm using chaotic initialization, a dynamic sweep mechanism, velocity-inspired updates, and an S-shaped binary transfer. Related work has also explored multi-strategy HOA adaptation [1], Q-learning and interaction-aware metaheuristics [7], and search-history-aware adaptation [5]. These studies establish that adaptive search and feature interaction are not new by themselves. The unresolved design question addressed here is narrower: can selection reproducibility close the loop around EHOA dynamics without inserting stability into the baseline fitness, and can the same evidence safely guide binary decisions?

The contributions are:

- a regime-aware feedback law in which exploration requires corroborating evidence of population collapse or stagnation, while late instability with healthy diversity causes consolidation;

- an entropy-confidence gate that suppresses ambiguous per-feature reliability evidence in the binary transition;
- a modular 2-by-2 ablation that leaves the EHOA wrapper objective untouched; and
- a leakage-safe, fully retained experiment record with predictive, reduction, stability, redundancy, cost, and non-parametric statistical evidence.

1.1 Research hypotheses and novelty boundary

The study tests three preregisterable, falsifiable hypotheses: H_1 , feedback increases chance-corrected subset stability relative to EHOA; H_2 , interaction guidance improves the held-out performance–reduction trade-off; and H_3 , the full method is Pareto non-dominated in balanced accuracy (maximize), Nogueira stability (maximize), and selected-feature count (minimize). Failure to improve accuracy alone does not confirm these hypotheses. The novelty claim is limited to the coupled mechanism: cross-resampling evidence controls continuous search dynamics through a corroborated two-regime law and controls discrete inclusion through a separate entropy-confidence gate. Neither adaptive metaheuristics nor relevance/redundancy scoring is claimed as novel in isolation.

2 Method

2.1 Baseline and notation

For N samples, D features, population P , and T iterations, continuous agent position $\mathbf{x}_i \in \mathbb{R}^D$ produces a binary mask $\mathbf{z}_i$. The unmodified baseline fitness is

$$f(\mathbf{z}) = \alpha(1 - \text{Acc}_{CV}(\mathbf{z})) + (1 - \alpha)\frac{\|\mathbf{z}\|_0}{D}, \quad \alpha = 0.99, \quad (1)$$

using deterministic 5-NN inside training-only stratified folds. DFA/SFIG-EHOA does not add stability or interaction terms to this objective.

2.2 Cross-resampling reliability

Class-stratified bootstrap samples are drawn only from the outer training set. Each resample produces a fixed-cardinality mutual-information ranking. Let $\hat{q}_j(t)$ be the fraction of resamples selecting feature j . Smoothed selection frequency and signed reliability are

$$q_j(t) = \rho q_j(t-1) + (1 - \rho)\hat{q}_j(t), \quad R_j(t) = 2q_j(t) - 1. \quad (2)$$

Global population stability is measured using both mean pairwise Jaccard and the chance-corrected Nogueira estimator [6]. Binary population diversity is normalized mean pairwise Hamming distance.

2.3 Regime-aware feedback

Let S_t be Nogueira stability clipped to $[0, 1]$, D_t binary diversity, $G_t \in [0, 1]$ stagnation, and $\tau = t/T$. Define uncertainty $U_t = 1 - S_t$, diversity ratio $r_t = \min(1, D_t/0.5)$, and collapse $L_t = 1 - r_t$. Exploration and consolidation drives are

$$E_t = L_t \left[\frac{1}{2}(1 - \tau) + \frac{1}{2}G_t \right], \quad C_t = U_t r_t \tau. \quad (3)$$

The signed controller adjustment is $A_t = \eta_E E_t - \eta_C C_t$. It modifies the baseline schedules as

$$SF_t = \text{clip}\{SF_t^{\text{base}}(1 + A_t), SF_{\min}, SF_{\max}\}, \quad (4)$$

$$\omega_t = \text{clip}\{\omega_t^{\text{base}} + 0.25A_t, \omega_{\min}, \omega_{\max}\}, \quad (5)$$

followed by exponential smoothing. Low stability alone cannot increase exploration. This is intentional: the initial additive-pressure prototype could create the loop “unstable $\rightarrow$ explore more $\rightarrow$ less stable.”

2.4 Confidence-gated interaction guidance

Binary entropy provides an evidence-confidence gate,

$$c_j(t) = 1 - H_2(q_j(t)). \quad (6)$$

Thus $q_j \approx 0.5$ contributes almost no directional force. Target mutual information supplies relevance. Cached absolute Pearson association to the current subset supplies a bounded redundancy term. This term is not claimed to be a higher-order interaction. The final transition is

$$p_{ij}(t) = \sigma [x_{ij}(t) + \lambda_s(t)c_j(t)R_j(t) + \lambda_i(t)I_j(S_i)]. \quad (7)$$

Weights support constant, linear, and signal-adaptive schedules. Four modes produce the required ablation: EHOA; SF-EHOA (stability feedback/reliability); IG-EHOA (interaction guidance); and DFA-EHOA (both). SFIG-EHOA is the descriptive alias for the full implementation.

2.5 Complexity

The wrapper remains dominant at approximately $O(TPCVC_{clf})$. Periodic stability adds resampling and mask comparison; the interaction model adds mutual information and a top- K correlation cache. Controller arithmetic is $O(T)$. Runtime and evaluation counts are measured rather than inferred because stochastic cache hits change effective cost.

3 Experimental Protocol

The study uses the versioned scikit-learn Breast Cancer Wisconsin Diagnostic (30 features, two classes) and Wine Recognition (13 features, three classes) datasets. Ten fixed seeds define paired 80/20 stratified hold-outs. Feature selection uses five-fold CV on the 80% training partition. Population size is 8 and every method receives 10 iterations. Reliability uses seven stratified resamples every two iterations, $\rho = 0.6$, $\lambda_s = \lambda_i = 0.5$, $\eta_E = 0.45$, and $\eta_C = 0.30$.

Imputation, scaling, and SMOTE [2] are fitted only within training folds or on the outer training partition. The test partition is never used for reliability, interaction, selection, or tuning. Balanced accuracy is primary; F1, MCC, ROC-AUC, PR-AUC, sensitivity, and specificity are secondary. Feature count, redundancy, interaction quality, Jaccard, Nogueira, runtime, classifier time, memory, and evaluations are retained.

Per dataset, a Friedman test compares the four paired variants. Wilcoxon signed-rank tests compare each extension with EHOA, with Holm correction, paired rank-biserial effect size, win/tie/loss counts, and a 5,000-resample bootstrap interval for the paired median difference. These tests follow common non-parametric comparison guidance [3]; with only two datasets, inference is reported per dataset and not generalized globally. Repeated nested CV was computationally unavailable. Repeated stratified hold-out is an explicitly weaker alternative and a limitation.

4 Results

Table 1: Held-out results over ten paired seeds. BA is balanced accuracy.

Dataset	Method	BA mean $\pm$ SD	Features	Nogueira	Runtime (s)
Breast	EHOA	0.9554 ± 0.0198	15.3	0.0322	2.99
Breast	SF-EHOA	0.9579 ± 0.0185	13.2	0.0891	9.12
Breast	IG-EHOA	0.9574 ± 0.0185	14.1	0.0499	9.49
Breast	DFA-EHOA	0.9498 ± 0.0215	13.9	0.1048	9.50
Wine	EHOA	0.9558 ± 0.0321	6.8	0.2188	2.02
Wine	SF-EHOA	0.9425 ± 0.0426	6.9	0.2073	5.75
Wine	IG-EHOA	0.9563 ± 0.0449	7.2	0.2667	5.71
Wine	DFA-EHOA	0.9558 ± 0.0321	6.9	0.2553	6.08

The full method improved Nogueira stability by 0.0726 on Breast Cancer and 0.0365 on Wine. On Breast Cancer it selected 1.4 fewer features while losing 0.0056 balanced accuracy. On Wine its mean balanced accuracy exactly matched EHOA and selected 0.1 more feature. DFA-EHOA ROC-AUC/PR-AUC was 0.9836/0.9832 on Breast Cancer and 0.9967/0.9921 on Wine. Jaccard and Nogueira do not rank every method identically because the latter corrects for chance and variable subset size; both outputs are therefore retained.

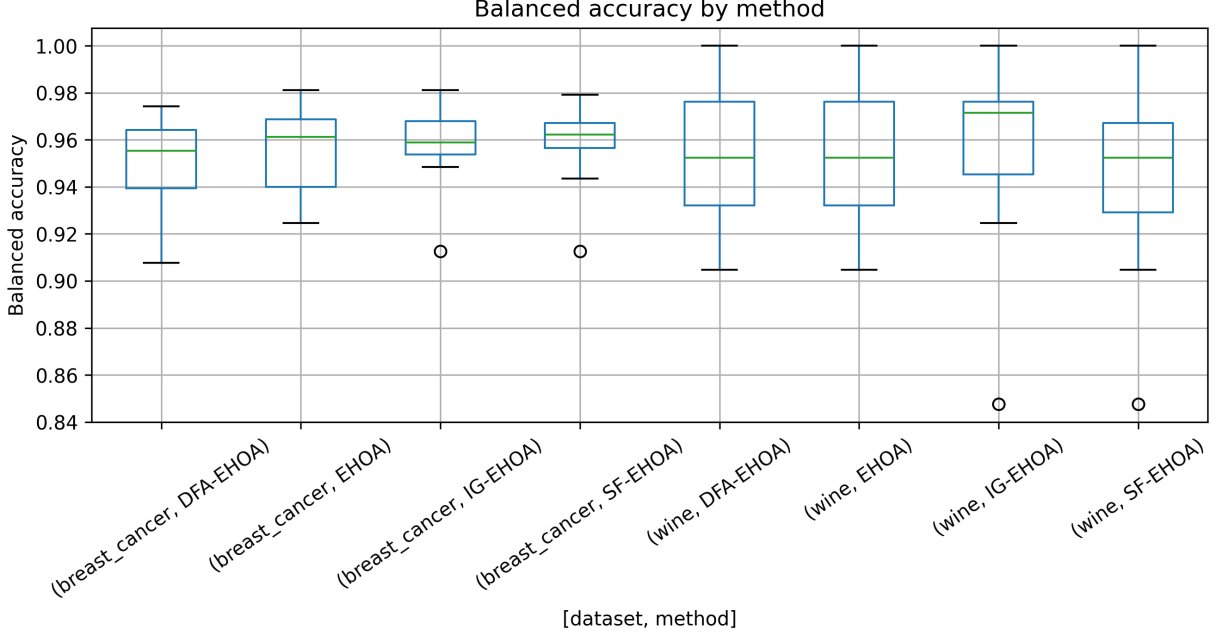


Figure 1: Balanced-accuracy distributions for the paired runs.

4.1 Ablation and telemetry

SF-EHOA obtained the best mean Breast Cancer balanced accuracy and smallest subset, but degraded Wine performance. IG-EHOA obtained the best Wine mean and stability, and the best Breast Cancer ROC-AUC/PR-AUC. The full combination prioritized chance-corrected reproducibility but did not combine component accuracy gains additively. This negative interaction is an important ablation result, not a failed run to omit.

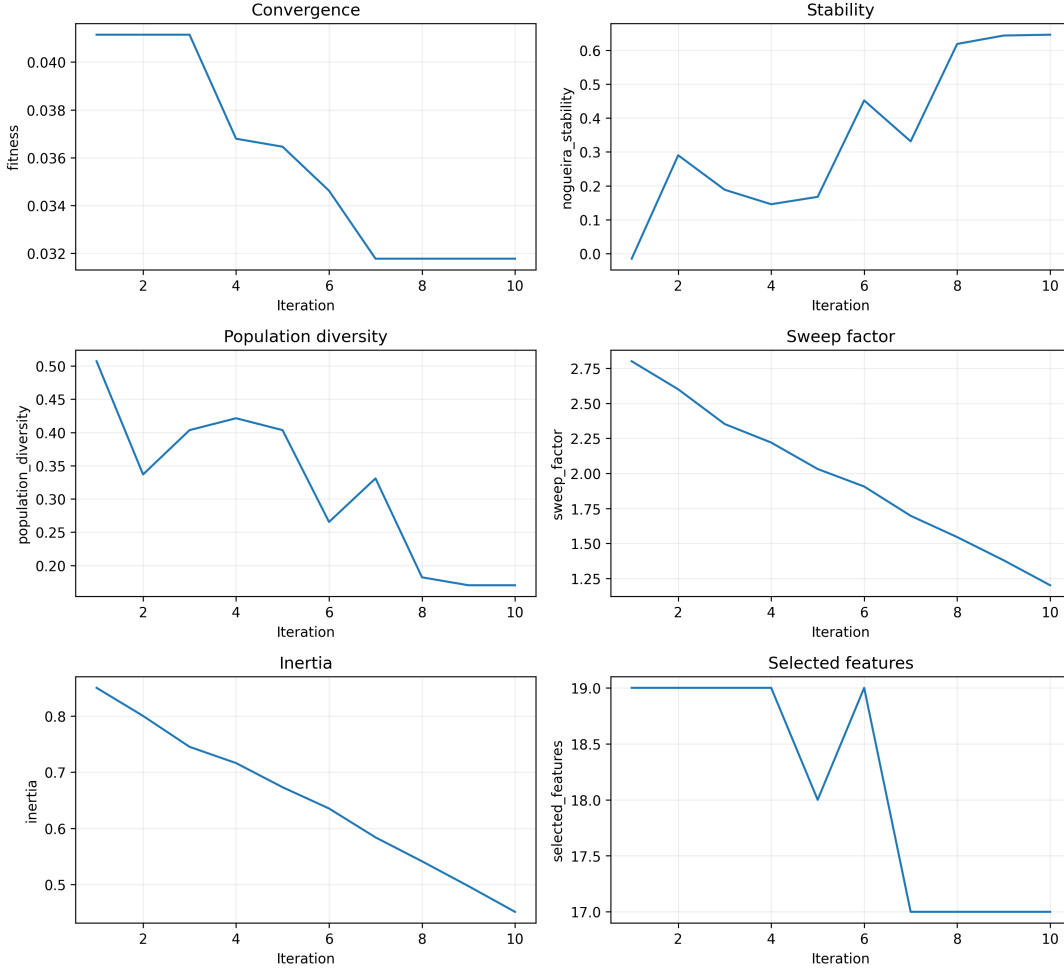


Figure 2: Representative full-method convergence and controller telemetry.

4.2 Statistical evidence and cost

Friedman p -values were 0.3039 for Breast Cancer and 0.0633 for Wine. For DFA-EHOA versus EHOA, Wilcoxon p -values were 0.2637 and 1.0000; all Holm-adjusted values were 1.0. Breast Cancer had a paired median change of -0.0060 (95% bootstrap interval -0.0169 to 0.0069), rank-biserial effect -0.418, and 3/0/7 wins/ties/losses. Wine had median change 0, interval 0 to 0, effect 0, and 1/8/1. No predictive-superiority claim is supported.

Full-method runtime was 3.18 times baseline on Breast Cancer and 3.01 times on Wine. The overhead is expected from resampling and interaction estimation and is not hidden by nominal same-iteration comparisons.

4.3 Pareto and hypothesis audit

A method is dominated only if another method is no worse on all three reported objectives and strictly better on at least one. DFA-EHOA is non-dominated on both datasets. On Breast Cancer, SF-EHOA has higher accuracy and fewer features, but lower Nogueira stability; hence neither dominates the other. On Wine, IG-EHOA has slightly higher accuracy and stability but uses more features, whereas EHOA uses fewer features with lower stability. DFA-EHOA also strictly dominates SF-EHOA on Wine. Accordingly, H_1 is supported descriptively for the full method on both datasets, H_2 receives component-level but

inconsistent support, and H_3 is supported for this limited benchmark. None is promoted to a population-wide or state-of-the-art claim because corrected predictive tests are non-significant and only two datasets were available.

5 Discussion

The experiment meets a scoped success criterion: on Breast Cancer, full DFA/SFIG-EHOA produced a meaningfully more chance-corrected stable and smaller subset with a small predictive penalty; on Wine, predictive performance was tied while stability improved. Whether this trade-off is desirable depends on the cost of biomarker inconsistency versus a half percentage point of balanced accuracy. The method is therefore useful as a controllable research mechanism, not established as a universally better optimizer.

The controller redesign is scientifically useful even without a predictive win. It rejects the simplistic assumption that instability always calls for more exploration. Separating early collapse recovery from late consolidation makes the response direction testable and prevents positive feedback.

6 Threats to Validity

Only two low-dimensional datasets were available in the supplied repository. Ten repeats provide paired evidence but low statistical power. Repeated hold-outs reuse observations and are weaker than repeated nested CV. The pairwise relevance/redundancy proxy cannot represent arbitrary synergistic interactions. Hyperparameters were fixed before the reported run, but the study lacks external validation. Runtime is platform-specific. These limitations prohibit state-of-the-art, medical, or universal claims.

7 Reproducibility and Conclusion

The repository contains the unchanged baseline, proposed module, 16 deterministic tests, YAML configuration, 80 raw runs, 80 traces, aggregate/statistical CSVs, figures, tables, a bilingual README, and an environment record. Runs are checkpointed and resumable. No failed seed was removed.

In conclusion, DFA/SFIG-EHOA contributes a real, ablatable stability-feedback and guided-transition design. Current evidence supports a stability–reduction trade-off, not predictive superiority. A pre-registered many-dataset study with repeated nested CV and evaluation-count-matched stopping is the appropriate next step.

Code and Data Availability

Source code and all reported artifacts are available in the public project repository. The exact CC BY 4.0 dataset snapshots are committed with source attribution and checksums; no private records are required. The software is MIT licensed under copyright 2026 Amirgh23.

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